

Consider the following instructions. Execute each one and write a report on the answers to the instructions. First complete the following tasks to create the database and the tables.

1. Create a database called “Bangla_Movies”. Write the corresponding query for this in your report.

2. Create a table titled “Movies” with the following constraints. Write the corresponding mySQL for this in your report.

```
CREATE TABLE Movies (  
    movie_id INT PRIMARY KEY,  
    title VARCHAR(100) NOT NULL,  
    release_year INT,  
    genre VARCHAR(50),  
    director_id INT,  
    box_office DECIMAL(12,2),  
    rating DECIMAL(3,1)  
);
```

3. Create a table titled “Directors” with the following constraints. Write the corresponding mySQL for this in your report.

```
CREATE TABLE Directors (  
    director_id INT PRIMARY KEY,  
    director_name VARCHAR(100) NOT NULL,  
    birth_year INT,  
    awards_count INT  
);
```

4. Create a table titled “Actors” with the following constraints. Write the corresponding mySQL for this in your report.

```
CREATE TABLE Actors (  
    actor_id INT PRIMARY KEY,  
    actor_name VARCHAR(100) NOT NULL,  
    birth_year INT,  
    awards_count INT  
);
```

5. Create a table titled “Movie_Actors” with the following constraints. Write the corresponding query for this in your report.

```
CREATE TABLE Movie_Actors (  
    movie_id INT,  
    actor_id INT,  
    role VARCHAR(100),  
    is_lead_role BOOLEAN,  
    PRIMARY KEY (movie_id, actor_id),  
    FOREIGN KEY (movie_id) REFERENCES Movies(movie_id),  
    FOREIGN KEY (actor_id) REFERENCES Actors(actor_id)  
);
```

6. Populate the table “Movies” with the following data. Write the corresponding insertion query for this in your report.

```
INSERT INTO Movies VALUES  
(1, 'No Dorai', 2023, 'Thriller', 1, 5.20, 8.5),  
(2, 'Poran', 2022, 'Romance', 4, 12.50, 7.8),  
(3, 'Shonibar Bikel', 2023, 'Horror', 4, 8.75, 8.1),  
(4, 'Rehana Maryam Noor', 2021, 'Drama', 5, 3.40, 8.7),  
(5, 'Gondi', 2022, 'Drama', 3, 4.80, 8.3),  
(6, 'Shimana Periyē', 2023, 'Romance', 2, 6.90, 7.5),  
(7, 'Mujib: The Making of a Nation', 2023, 'Biography', NULL, 15.20, 8.9),  
(8, 'Debi', 2023, 'Horror', 2, 7.30, 7.2),  
(9, 'Dui Diner Duniya', 2022, 'Comedy', 1, 5.60, 7.0),  
(10, 'Kura Pokkhir Shunne Ura', 2023, 'Drama', 3, 2.80, 8.4);
```

7. Populate the table “Movie_Actors” with the following data. Write the corresponding insertion query for this in your report.

```
INSERT INTO Movie_Actors VALUES  
(1, 1, 'Protagonist', TRUE),  
(1, 2, 'Supporting', FALSE),  
(2, 7, 'Lead Male', TRUE),  
(2, 8, 'Lead Female', TRUE),  
(3, 3, 'Protagonist', TRUE),  
(3, 4, 'Supporting', FALSE),  
(4, 6, 'Rehana Maryam Noor', TRUE),  
(5, 1, 'Protagonist', TRUE),  
(5, 5, 'Supporting', FALSE),  
(6, 7, 'Lead Male', TRUE),  
(6, 9, 'Lead Female', TRUE),  
(7, 1, 'Bangabandhu Sheikh Mujibur Rahman', TRUE),
```

```
(7, 10, 'Supporting', FALSE),  
(8, 3, 'Protagonist', TRUE),  
(8, 5, 'Supporting', FALSE),  
(9, 5, 'Protagonist', TRUE),  
(9, 2, 'Supporting', FALSE),  
(10, 4, 'Protagonist', TRUE),  
(10, 6, 'Supporting', FALSE);
```

7. Populate the table “Actors” with the following data. Write the corresponding insertion query for this in your report.

```
INSERT INTO Actors VALUES  
(1, 'Chanchal Chowdhury', 1974, 20),  
(2, 'Fazlur Rahman Babu', 1964, 12),  
(3, 'Jaya Ahsan', 1978, 18),  
(4, 'Shahiduzzaman Selim', 1965, 10),  
(5, 'Mosharraf Karim', 1972, 15),  
(6, 'Nusrat Imrose Tisha', 1985, 8),  
(7, 'Siam Ahmed', 1990, 5),  
(8, 'Puja Cherry Roy', 1995, 3),  
(9, 'Tasnia Farin', 1992, 4),  
(10, 'Shatabdi Wadud', 1980, 6);
```

8. Populate the table “Directors” with the following data. Write the corresponding insertion query for this in your report.

```
INSERT INTO Directors VALUES  
(1, 'Mostofa Sarwar Farooki', 1973, 15),  
(2, 'Tauquir Ahmed', 1965, 12),  
(3, 'Giasuddin Selim', 1964, 8),  
(4, 'Nuhash Humayun', 1993, 5),  
(5, 'Rubaiyat Hossain', 1981, 7);
```

Queries and solutions:

- a) List all movies released in 2023 with their ratings.

```
SELECT title, rating  
FROM Movies  
WHERE release_year = 2023;
```

→T←				<u>title</u>	<u>rating</u>
☐	Edit	Copy	Delete	No Dorai	8.5
☐	Edit	Copy	Delete	Shonibar Bikel	8.1
☐	Edit	Copy	Delete	Shimana Periyē	7.5
☐	Edit	Copy	Delete	Mujib: The Making of a Nation	8.9
☐	Edit	Copy	Delete	Debi	7.2
☐	Edit	Copy	Delete	Kura Pokkhir Shunne Ura	8.4

b) Find all actors who have won more than 10 awards.

```
SELECT actor_name, awards_count
FROM Actors
WHERE awards_count > 10;
```

←T→				<u>actor_name</u>	<u>awards_count</u>
☐	Edit	Copy	Delete	Chanchal Chowdhury	20
☐	Edit	Copy	Delete	Fazlur Rahman Babu	12
☐	Edit	Copy	Delete	Jaya Ahsan	18
☐	Edit	Copy	Delete	Mosharraf Karim	15

c) Show movies with box office collection greater than 5 crore and rating above 8.

```
SELECT title, box_office, rating
FROM Movies
WHERE box_office > 5 AND rating > 8;
```

←T→				<u>title</u>	<u>box_office</u>	<u>rating</u>
☐	Edit	Copy	Delete	No Dorai	5.20	8.5
☐	Edit	Copy	Delete	Shonibar Bikel	8.75	8.1
☐	Edit	Copy	Delete	Mujib: The Making of a Nation	15.20	8.9

d) List all horror movies with their directors.

```
SELECT m.title, d.director_name
FROM Movies m
JOIN Directors d ON m.director_id = d.director_id
WHERE m.genre = 'Horror';
```

<u>title</u>	<u>director_name</u>
Shonibar Bikel	Nuhash Humayun
Debi	Tauquir Ahmed

e) Find all lead actors in movies directed by Nuhash Humayun.

```
SELECT a.actor_name, m.title
FROM Actors a
JOIN Movie_Actors ma ON a.actor_id = ma.actor_id
JOIN Movies m ON ma.movie_id = m.movie_id
JOIN Directors d ON m.director_id = d.director_id
WHERE d.director_name = 'Nuhash Humayun' AND ma.is_lead_role = TRUE;
```

<u>actor_name</u>	<u>title</u>
Siam Ahmed	Poran
Puja Cherry Roy	Poran
Jaya Ahsan	Shonibar Bikel

f) List all movies where Chanchal Chowdhury is an actor (lead or supporting)

```
SELECT m.title, ma.role,
CASE WHEN ma.is_lead_role THEN 'Lead' ELSE 'Supporting' END as role_type
FROM Movies m
JOIN Movie_Actors ma ON m.movie_id = ma.movie_id
JOIN Actors a ON ma.actor_id = a.actor_id
WHERE a.actor_name = 'Chanchal Chowdhury';
```

<u>actor_name</u>	<u>title</u>
Siam Ahmed	Poran
Puja Cherry Roy	Poran
Jaya Ahsan	Shonibar Bikel

g) Show directors who have directed movies in multiple genres.

```
SELECT d.director_name, COUNT(DISTINCT m.genre) as genres_count
FROM Directors d
JOIN Movies m ON d.director_id = m.director_id
GROUP BY d.director_id, d.director_name
HAVING COUNT(DISTINCT m.genre) > 1;
```

<u>director_name</u>	<u>genres_count</u>
Mostofa Sarwar Farooki	2
Tauquir Ahmed	2
Nuhash Humayun	2

h) Find actors who have worked with director Mostofa Sarwar Farooki.

```
SELECT DISTINCT a.actor_name
FROM Actors a
JOIN Movie_Actors ma ON a.actor_id = ma.actor_id
JOIN Movies m ON ma.movie_id = m.movie_id
JOIN Directors d ON m.director_id = d.director_id
WHERE d.director_name = 'Mostofa Sarwar Farooki';
```

<u>actor_name</u>
Chanchal Chowdhury
Fazlur Rahman Babu
Mosharraf Karim

i) List movies that don't have a director assigned.

```
SELECT m.title, m.release_year
FROM Movies m
LEFT JOIN Directors d ON m.director_id = d.director_id
WHERE d.director_id IS NULL;
```

title	release_year
Mujib: The Making of a Nation	2023

j) Find all romance movies released after 2021 along with their lead actors' names.

```
SELECT m.title, m.release_year, a.actor_name
FROM Movies m
JOIN Movie_Actors ma ON m.movie_id = ma.movie_id
JOIN Actors a ON ma.actor_id = a.actor_id
WHERE m.genre = 'Romance'
AND m.release_year > 2021
AND ma.is_lead_role = TRUE;
```

<u>title</u>	<u>release_year</u>	<u>actor_name</u>
Poran	2022	Siam Ahmed
Poran	2022	Puja Cherry Roy
Shimana Periye	2023	Siam Ahmed
Shimana Periye	2023	Tasnia Farin

****THE END****